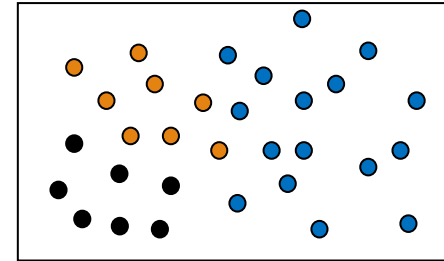


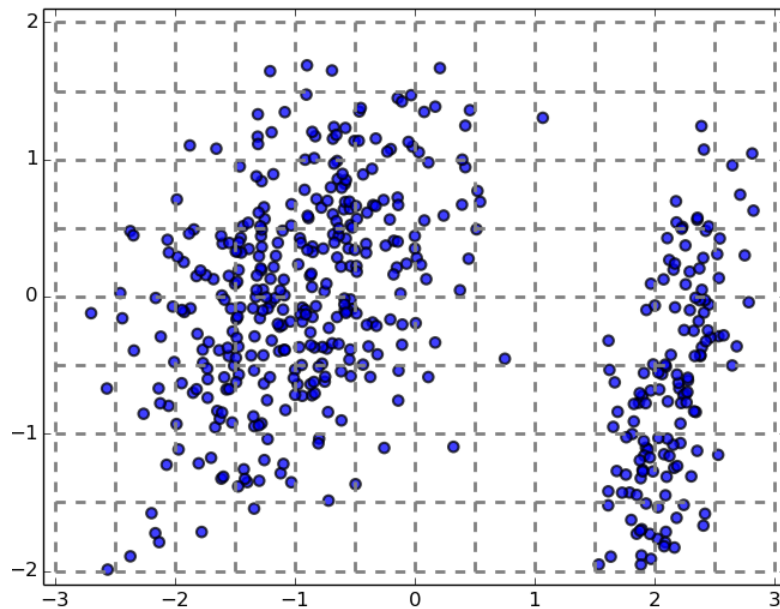
Clustering Tendency: Whether the Data Contains Inherent Grouping Structure

- ❑ Assess the suitability of clustering
 - ❑ Whether the data has any “inherent grouping structure” — non-random structure that may lead to meaningful clusters
- ❑ Determine clustering tendency or clusterability
 - ❑ A hard task because there are so many different definitions of clusters
 - ❑ Different definitions: Partitioning, hierarchical, density-based and graph-based
 - ❑ Even fixing a type, still hard to define an appropriate null model for a data set
- ❑ There are some clusterability assessment methods, such as
 - ❑ Spatial histogram: Contrast the histogram of the data with that generated from random samples
 - ❑ Distance distribution: Compare the pairwise point distance from the data with those from the randomly generated samples
 - ❑ Hopkins Statistic: A sparse sampling test for spatial randomness

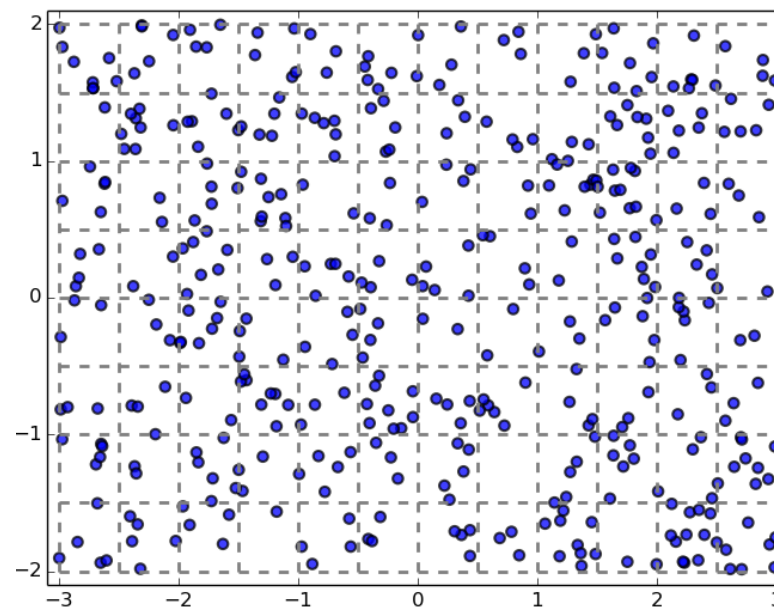


Testing Clustering Tendency: A Spatial Histogram Approach

- ❑ Spatial Histogram Approach: Contrast the d-dimensional histogram of the input dataset D with the histogram generated from random samples
- ❑ Dataset D is clusterable if the distributions of two histograms are rather different



(a) Input dataset



(b) Data generated from random samples

Testing Clustering Tendency: A Spatial Histogram Approach

- ❑ Method outline
 - ❑ Divide each dimension into equi-width bins, count how many points lie in each cell, and obtain the empirical joint probability mass function (EPMF)
- ❑ Do the same for the randomly sampled data
- ❑ Compute how much they differ using the Kullback-Leibler (KL) divergence value